

Chapter 2 Notes: Block Diagrams

1. Why Block Diagrams?

A real control system is made of multiple devices (sensors, effectors, plant), each represented mathematically by a **transfer function**. These

For a single device, this is trivial:

$$U(s) \rightarrow G(s) \rightarrow Y(s) \quad \Rightarrow \quad Y(s) = G(s)U(s)$$

But once you have several connected devices, tracking the algebra by hand gets messy. A **block diagram** is a visual + mathematical tool that

Internal vs. external signals

$$U(s) \rightarrow G_1(s) \rightarrow X(s)$$

$$X(s) \rightarrow G_2(s) \rightarrow Y(s)$$

$U(s)$ and $Y(s)$ are **external signals** (the overall input/output).

$X(s)$ — the signal between the two blocks — is an **internal signal**.

Solving it:

$$Y(s) = G_2(s)X(s), \quad X(s) = G_1(s)U(s)$$

$$\Rightarrow Y(s) = G_2(s)G_1(s)U(s) \quad \Rightarrow \quad U(s)Y(s) = G_1(s)G_2(s) \quad (2.1)$$

Takeaway: cascaded (chained) transfer functions simply **multiply**.

2. Three Fundamental Building Blocks

(a) Transfer Function

A linear, time-invariant differential equation written in the Laplace domain as output/input, valid only when **all initial conditions are zero**.

(b) Summation Junction

Adds (or subtracts) multiple input signals into one output signal:

$$X(s) = \sum_{i=1}^N U_i(s) \quad (2.2)$$

Drawn as a small circle with several signals flowing in and one flowing out.

A **minus sign** next to an incoming arrow means that signal is *subtracted*.

No sign shown = positive (implied $\boxed{+}$).

Example: if $U_1(s)$ enters normally and $U_N(s)$ enters with a minus sign, then $X(s) = U_1(s) - U_N(s)$.

Cascading summation junctions: Multiple summing junctions in a row can always be collapsed into **one** summing junction that adds all the

$$Y(s) = U_1(s) + U_2(s) + U_3(s) + U_4(s)$$

****Real-world example — Fahrenheit to Celsius conversion:**** A summation junction (subtracting 32) followed by a gain block (5/9) represents

$$\text{Celsius} = 9/5(\text{Fahrenheit} - 32)$$

This shows block diagrams aren't just abstract circuit theory — they can represent *any* mathematical process.

(c) Branch

An **ideal split** — a signal splits into two (or more) identical copies, each following its own path. No signal loss or attenuation happens at a branch.

3. Section 2.1 — Manipulation: The Standard Feedback Model

The single most important diagram in this chapter — a system with **negative feedback**:

$$U(s) \rightarrow \bigcirc(-) \rightarrow G(s) \rightarrow Y(s), \quad \text{with } Y(s) \text{ fed back through } H(s) \text{ into the summing junction}$$

$$G(s) = \text{forward path transfer function}$$

$$H(s) = \text{feedback (sensor) transfer function}$$

Derivation:

$$Y(s) = G(s)E(s), \quad E(s) = U(s) - H(s)Y(s)$$

$$Y(s) = G(s)(U(s) - H(s)Y(s)) = G(s)U(s) - G(s)H(s)Y(s)$$

$$Y(s) + GH(s)Y(s) = GU(s) \Rightarrow (1 + GH(s))Y(s) = GU(s)$$

$$U(s)Y(s) = 1 + GH(s)G(s) \quad (2.3, 2.5)$$

Memorize this formula — it's the classic negative-feedback transfer function and shows up constantly.

Continued-fraction form

Dividing numerator and denominator by the denominator gives an alternative, useful way to write (2.3):

$$T(s) = 1 + GH(s)G(s) = G(s)1 + H(s)1 \quad (2.4)$$

Why this matters — going *backwards* from a transfer function to a block diagram: Given a transfer function like

$$T(s) = s^2 + a_1s + a_0s + b_0$$

you can rewrite it in continued-fraction form and match term-by-term against (2.4) to identify $G(s)$ and $H(s)$:

$$\text{Rewrite: } T(s) = \frac{1}{\frac{s^2 + a_1s}{s + b_0} + \frac{a_0}{s + b_0}} T(s) = s + b_0 \frac{s^2 + a_1s + a_0}{s + b_0}$$

$$\text{Match to } \frac{1}{1/G(s) + H(s)}$$

Rule of thumb: choose the lowest-order expression for $H(s)$. Here that means $H(s) = \frac{a_0}{s + b_0}$, leaving $\frac{1}{G(s)} = \frac{s^2 + a_1s + a_0}{s + b_0}$

⚠ Picking the wrong split for GG/HH still gives a mathematically valid model, but it won't behave correctly in simulation software — always

4. Worked Examples (Section 2.1)

Example — Cascaded Blocks

$$\text{Given } \frac{4}{s + 4} \text{ cascaded into } \frac{s + 1}{s^2 + 2s + 1} \Rightarrow Y(s) = \frac{4(s + 1)}{(s + 4)(s^2 + 2s + 1)} U(s)$$

$$Y(s) = \frac{4(s + 1)}{(s + 4)(s^2 + 2s + 1)} U(s) = \frac{4s^3 + 6s^2 + 9s + 4}{(s + 4)(s^2 + 2s + 1)} U(s)$$

$$U(s)Y(s) = 4s^3 + 6s^2 + 9s + 4$$

Convention: always write the final transfer function as a ratio of **descending-order polynomials**.

Example — Negative Feedback (general form)

Re-derives (2.5) directly from the diagram — same result as Section 3 above:

$$U(s)Y(s) = 1 + GH(s)$$

Example — Negative Feedback (numeric)

$$\text{Given } G(s) = \frac{13}{s + 13}, H(s) = \frac{4}{s + 4} \Rightarrow Y(s) = \frac{13}{s + 13} \frac{4}{s + 4} U(s) = \frac{52}{(s + 13)(s + 4)} U(s)$$

$$Y(s) = \frac{52}{(s + 13)(s + 4)} U(s)$$

Distribute, collect $Y(s)$ terms, and simplify step by step:

$$Y(s) = \frac{52}{(s + 13)(s + 4)} U(s) \Rightarrow (1 + \frac{52}{(s + 13)(s + 4)}) Y(s) = \frac{52}{(s + 13)(s + 4)} U(s)$$

$$(1 + \frac{52}{(s + 13)(s + 4)}) Y(s) = \frac{52}{(s + 13)(s + 4)} U(s)$$

$$Y(s) = ((s + 13)(s + 4) + 52(s + 13)(s + 4))s + 1313U(s) = ((s + 13)(s + 4) + 5213(s + 4))U(s)$$

Final transfer function:

$$U(s)Y(s) = s^2 + 17s + 10413s + 52$$

Example — Feedforward

Two blocks $G_1(s)$ and $G_2(s)$ both take the *same* input $U(s)$, and their outputs are summed:

$$Y(s) = G_1U(s) + G_2U(s) = (G_1(s) + G_2(s))U(s)$$

$$U(s)Y(s) = G_1(s) + G_2(s) \quad (2.6)$$

****Takeaway:**** parallel paths feeding into a summing junction ****add****.

5. Section 2.2 — Block Diagram Algebra

Several standard manipulations let you redraw a block diagram into an equivalent, often simpler, form without changing its input/output behavior.

2.2.1 Cascaded Block Manipulation

A chain of transfer functions can be regrouped in many equivalent ways, as long as $Y(s)/U(s)$ stays the same. Example: $\frac{4}{s+4}s + 44$

$$\begin{aligned} &\text{One block } \frac{4s + 4}{s^3 + 6s^2 + 9s + 4} s^3 + 6s^2 + 9s + 44s + 4, \text{ or} \\ &\frac{1}{s^3 + 6s^2 + 9s + 4} s^3 + 6s^2 + 9s + 41 \text{ then } 4s + 44s + 4, \text{ or} \\ &44 \text{ then } \frac{1}{s^3 + 6s^2 + 9s + 4} s^3 + 6s^2 + 9s + 41 \text{ then } s + 1s + 1 \end{aligned}$$

All are mathematically equivalent — how you split the numerator/denominator across blocks is a matter of representation.

2.2.2 Moving Summation Junctions

Moving a summation junction *after* a transfer function: Original: $U_1(s)$ and $-U_2(s)$ summed, then through $G(s)$:

$$Y = G(s)(U_1(s) - U_2(s)) = GU_1(s) - GU_2(s) \quad (2.7)$$

To move the junction after $G(s)$, route U_1 through its own $G(s)$ block and U_2 through an unknown $H(s)$ block, then sum:

$$Y = GU_1(s) - HU_2(s) \quad (2.8)$$

Comparing (2.7) and (2.8): $H(s) = G(s)$. **So when moving a junction *after* a transfer function, add a matching copy of that transfer function.**

Moving a summation junction *before* a transfer function: Original: $Y(s) = GU_1(s) - U_2(s)$ (2.9). Moving the junction

2.2.3 Moving Branches

Moving a branch *after* a transfer function: insert $H(s) = \frac{1}{G(s)}$ into the moved branch (since $U_1(s) = \frac{1}{G(s)}Y(s)$ $U_1(s) = G(s)^{-1}Y(s)$)

Moving a branch *before* a transfer function: insert $H(s) = G(s)$ into the moved branch (since $Y(s) = GU(s)$ $Y(s) = GU(s)$ directly)

Pattern to remember:

Moving...	...past a transfer function $G(s)$	Insert into the moved path
Summation junction after $G(s)$		$G(s)G(s)$
Summation junction before $G(s)$		$1/G(s)1/G(s)$
Branch after $G(s)$		$1/G(s)1/G(s)$
Branch before $G(s)$		$G(s)G(s)$

2.2.4 Combining Parallel Paths

Same idea as the Feedforward example above: if $G_1(s)G_1(s)$ and $G_2(s)G_2(s)$ share the same input and their outputs are summed,

$$Y(s) = G_1U(s) + G_2U(s) \Rightarrow U(s)Y(s) = G_1(s) + G_2(s)$$

Parallel paths into a summing junction **combine by addition**.

2.2.5 Eliminating a Feedback Loop

General form allowing **either** positive or negative feedback (the $\pm \pm / \mp \mp$ signs track together):

$$Y(s) = G(s)E(s), \quad E(s) = U(s) \pm H(s)Y(s)$$

$$Y(s) = GU(s) \pm GHY(s) \Rightarrow Y(s) \mp GHY(s) = GU(s) \Rightarrow (1 \mp GH(s))Y(s) = GU(s)$$

$$U(s)Y(s) = 1 \mp GH(s)G(s)$$

(Negative feedback $\rightarrow$ denominator is $1+GH(s)$; positive feedback $\rightarrow$ denominator is $1-GH(s)$.)

6. Open-Loop vs. Closed-Loop

Take a standard feedback diagram with input $R(s)$ (reference), output $Y(s)$, forward path $G(s)$, negative feedback $H(s)$:

$$R(s)Y(s) = 1 + GH(s)G(s)$$

Closed-loop system: has a feedback loop — the signal can loop back on itself. The closed-loop transfer function is $\frac{G(s)}{1 + GH(s)}$

Open-loop system: no feedback path — the signal flows straight from input to output. Every closed-loop system can be algebraically simplified to an open-loop system.

Loop-gain $L(s)$: the product of all transfer functions *within* the loop. In the example above, $L(s) = G(s)H(s)$.

Why both representations matter:

The **open-loop** form $\frac{Y(s)}{R(s)} = \frac{B(s)}{A(s)}$ — the denominator $A(s)$ is called the **characteristic polynomial** and determines system stability.

The **closed-loop** form $\frac{G(s)}{1 + L(s)}$ exposes the loop-gain $L(s)$, which governs response error and frequency-domain stability behavior.

You'll often need **both** views depending on what you're analyzing.

Worked Example — Open-Loop $\rightleftharpoons$ Closed-Loop Conversion

Given: $G(s) = \frac{s + 13}{s^2 + 5s + 13}$, $H(s) = 1$ (unity feedback), negative feedback.

Closed-loop form:

$$R(s)Y(s) = 1 + GH(s)G(s) = 1 + s^2 + 5s + 13s + 13s^2 + 5s + 13s + 13$$

Simplify to get the open-loop-style ratio of polynomials:

$$R(s)Y(s) = (s^2 + 5s + 13) + (s + 13)s + 13 = s^2 + 6s + 26s + 13$$

So:

closed-loop form

$$1 + s^2 + 5s + 13s + 13s^2 + 5s + 13s + 13 \Leftrightarrow \text{open-loop-style / simplified form}$$

$$s^2 + 6s + 26s + 13$$

These are algebraically equivalent — same transfer function — but each form highlights different things about the system's behavior.

Quick-Reference Summary

Cascaded blocks multiply: $Y/U = G_1G_2 \Rightarrow Y/U = G_1G_2$.

Parallel/feedforward blocks add: $Y/U = G_1 + G_2 \Rightarrow Y/U = G_1 + G_2$.

Negative feedback formula (memorize): $\frac{Y(s)}{U(s)} = \frac{G(s)}{1 + GH(s)}$; for positive feedback, flip the sign to $1 - GH(s)$.

Continued-fraction trick: lets you go *backwards* from a transfer function to a feedback block diagram — always pick the lowest-order $H(s)$.

Moving a junction/branch across a block: after $\rightarrow$ multiply by $G(s)$ (junction) or by $1/G(s)$ (branch); before $\rightarrow$ the opposite. (See

Open-loop vs. closed-loop: same system, two equivalent representations. Open-loop's denominator = characteristic polynomial (stability);

Final transfer functions are conventionally written as a ratio of **descending-order polynomials**.

(Note: Section 2.6 was skipped per your request.)